

Tarea dección 7.2

9) $\alpha = 27^\circ 13'$ $a = 1.10$

$\beta = 43^\circ 40'$ $b = 2.10$

$\gamma = 115^\circ 10'$ $c = 2.75$

$$c^2 = 2.10^2 + 1.10^2 - 2(1.10)(2.10)\cos 115^\circ 10'$$

$$c^2 = 7.59$$

$$c = \sqrt{7.59} = 2.75$$

$$\frac{\sin 115^\circ 10'}{2.75} = \frac{\sin \beta}{2.10} \Rightarrow \sin^{-1}\left(\frac{2.10 \sin 115^\circ 10'}{2.75}\right) = \beta = 43^\circ 40'$$

$$\frac{\sin 115^\circ 10'}{2.75} = \frac{\sin \alpha}{1.10} \Rightarrow \sin^{-1}\left(\frac{1.10 \sin 115^\circ 10'}{2.75}\right) = \alpha = 27^\circ 13'$$

17) $\alpha = 89$ $a = 286.5$

$\beta = 89$ $b = 286.5$

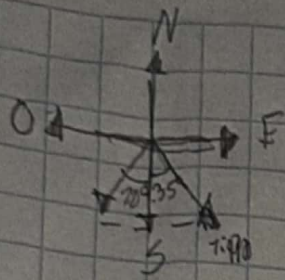
$\gamma = 2$ $c = 10.0$

$$\gamma = \cos^{-1}\left(\frac{10^2 - 286.5^2 - 286.5^2}{2(286.5)(10)}\right) = \gamma = 7.99 \approx 2$$

$$\beta = \cos^{-1}\left(\frac{286.5^2 + 10^2 - 286.5^2}{2(286.5)(10)}\right) = \beta = 89$$

$$\alpha = 180 - 89 - 2 = 89$$

23)



$$24/2 = 48 \text{ mi}$$

$$20/35 = 55$$

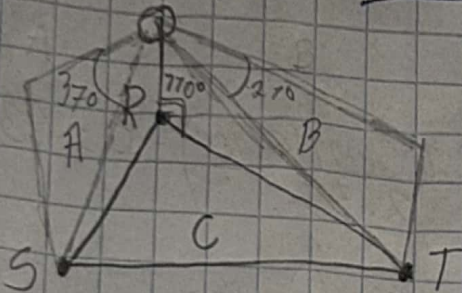
$$18/1.5 = 27 \text{ mi}$$

$$a^2 = 27^2 + 48^2 - 2(27)(48) \cos 55$$

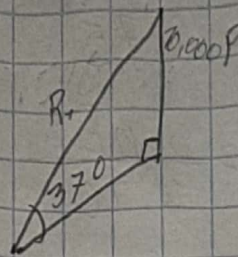
$$a = \sqrt{9,546.29}$$

$$a = 39 \text{ millas}$$

37)

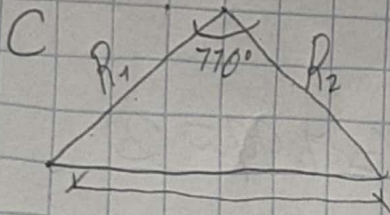
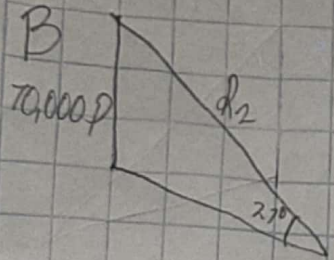


A



$$\sin 37^\circ = \frac{10,000}{R_1} \Rightarrow R_1 = 16,676.40$$

$$\sin 27^\circ = \frac{10,000}{R_2} \Rightarrow R_2 = 27,904.28$$

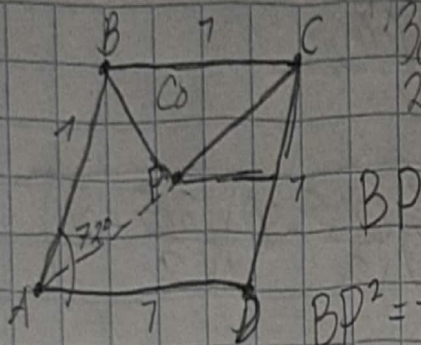


$$ST^2 = 16,676.40^2 + 27,904.28^2 - 2(16,676.40)(27,904.28) \cos 77^\circ$$

$$ST = \sqrt{7,377,927,647}$$

ST = 37,039.46 m la distancia entre el submarino y el buque mercante

35)



$$360 - 72 - 72 = 216$$

$$216 / 2 = 108, \text{ cada angulo}$$

$$BD^2 = 7^2 + 7^2 - 2(7)(7)\cos 36$$

$$BP = 0.62$$

$$\frac{\sin 36}{0.61} = \frac{\sin 111.8}{1}$$

$$BPC = 180 - 74.49 - 36$$

$$BPC = 69.51$$

$$\angle B = 74.49$$

$$APB = 180 - 69.51$$

$$APB = 110.49$$

$$ABD = 180 - 74.49$$

$$ABP = 105.51$$

36) $\alpha = 40.3^\circ$

$\beta = 62.9^\circ$ $b = 5.63$

$$\frac{\sin 62.9}{5.65} = \frac{\sin 40.5}{a} = a = 4.09$$

$$\frac{\sin 40.5}{4.09} = \frac{\sin 76.8}{c} = c = 6.76$$

$$s = \frac{1}{2}(4.09 + 5.65 + 6.76) = 7.94$$

$$A = \sqrt{7.94(7.94 - 4.09)(7.94 - 5.63)(7.94 - 6.76)}$$

$$A = 11.27$$

43)

$$a = 25$$

$$b = 80$$

$$c = 60$$

$$s = \frac{1}{2}(25 + 80 + 60)$$

$$s = 765/2$$

$$A = \sqrt{\frac{765}{2} \left(\frac{765}{2} - 25 \right) \left(\frac{765}{2} - 80 \right) \left(\frac{765}{2} - 60 \right)}$$

$$A = 577$$